Jian Hazel Sitorus

Informatics Undergraduate | Data Science and Web Development

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PROFESSIONAL PROFILE

Informatics undergraduate at Telkom University with experience in applied data science research, machine learning, and web development. Work includes a systematic literature review of tourism route recommendation systems, a multi-day Jakarta itinerary recommendation application, predictive modeling projects, and a multi-vendor marketplace. Uses Python for data analysis and modeling, with Laravel, Vue.js, React, and FastAPI across web projects.

EDUCATION

Informatics Undergraduate 09/2023 – 04/2027
Telkom University Bandung, West Java
GPA: 3.6/4.00 Expected graduation: April 2027.

PROFESSIONAL EXPERIENCE

Web Developer Intern 08/2026 – 10/2026
PT PLN (Persero) - UPDL Suralaya Pulo Merak, Banten, Indonesia (Hybrid)

- Developed OneSpace, an internal room, dormitory, and facility booking system for PLN UPDL Suralaya using Laravel 12, Inertia, React, TypeScript, and PostgreSQL.
- Enforced double-booking prevention with PostgreSQL exclusion constraints over time ranges to address race conditions.
- Developed automated tests for booking and approval rules.
- Prepared the application for production deployment with CSP nonces, trusted proxy handling, secure session cookies, and a documented production checklist.

Data Scientist Intern 04/2026 – 06/2026
HUMIC Engineering Research Bandung, West Java

- Conducted a systematic literature review of optimization-based tourism route recommendation systems, screening 135 papers and selecting 67 using the Parsifal platform.
- Extracted algorithms, datasets, constraints, objective functions, and evaluation metrics from the 67 selected papers.
- Developed a research taxonomy in Whimsical covering nine dimensions: objective, methodology, advantages, disadvantages, model, metric, contribution, future work, and dataset.
- Built a multi-day Jakarta tourist itinerary recommendation system using Content-Based Filtering, K-Means clustering, and hybrid GWO-TS optimization on the Massive-STEPS Jakarta dataset.
- Deployed the end-to-end prototype as a web application and evaluated fitness value, running time, and convergence.

TECHNICAL SKILLS

Programming Languages
Python, SQL, PHP, JavaScript, TypeScript

Web Frameworks and Libraries
Laravel, Vue.js, React, FastAPI, Tailwind CSS

Data Science Libraries
Pandas, NumPy, scikit-learn, Matplotlib, Seaborn

Tools and Database
Git, GitHub, VS Code, Jupyter Notebook, MySQL, PostgreSQL

HONORS AND AWARDS

9th Place
Hackathon: Retail Crisis & Recovery Visualization Challenge using Python

TECHNICAL PROJECTS

Jakarta Routes Multi-Day Tourist Itinerary Recommendation System
Technologies: Python, FastAPI, React, Vite, Tailwind CSS, scikit-learn, Pandas, NumPy, Jupyter Notebook, Leaflet, Vercel
- Developed a full-stack recommendation system for personalized multi-day tourist itineraries in Jakarta.
- Implemented Content-Based Filtering with TF-IDF and geographic clustering to recommend destinations based on user preferences.
- Integrated GWO-TS route optimization while considering travel time, opening hours, and daily time constraints.
- Processed tourism data from OpenStreetMap, Google Places, Wikipedia, and real-world check-in datasets; built an interactive Leaflet map and deployed the frontend and backend on Vercel.

Blukios Multi-Vendor Marketplace Platform
Technologies: Laravel, Vue.js, PHP, Python, MySQL, MongoDB
- Developed a multi-vendor marketplace with a Laravel backend and Vue.js interface.

Ames House Price Prediction
Technologies: Python, Pandas, NumPy, scikit-learn, XGBoost, Matplotlib, Seaborn
- Prepared Ames Housing data with missing-value handling and engineered TotalSF and HouseAge features.
- Trained an XGBoost regressor and explored hyperparameters using five-fold cross-validation; evaluated predictions with MAE and RMSE.

Insurance Cost Prediction
Technologies: Python, NumPy, Pandas, Matplotlib, Jupyter Notebook
- Implemented Linear Regression from scratch and explored a Genetic Algorithm for parameter optimization in an insurance cost prediction project.

Credit Risk Prediction
Technologies: Python, Pandas, NumPy, scikit-learn, XGBoost, Matplotlib, Seaborn
- Built an academic credit-risk classification project using Lending Club data; compared Logistic Regression, XGBoost, and a stacking ensemble, and explored precision-recall trade-offs.

LEADERSHIP AND ACTIVITIES

Hello World Telkom University 06/2024 – 09/2024
Logistics Committee Member Bandung, West Java
- Coordinated event logistics, managing supplies and consumption needs.
- Collaborated with team members on timely resource delivery and efficient allocation; helped resolve logistical issues during the event.

PMK Telkom University 12/2023 – 12/2023
Logistics Committee Member | Christmas Committee Bandung, West Java
- Organized logistical preparations for the Christmas celebration, including material procurement and event setup.
- Contributed to financial goals supporting the overall success of the celebration.

CERTIFICATIONS AND TRAINING

Data Science Essentials With Python
Cisco

Data Scientist Project Based Internship Program
ID/X Partners

Learn the Basics of Git with GitHub
Dicoding

Getting Started with Python Programming
Dicoding

Prompt Engineering for Software Developers
Dicoding

Web Programming Laravel
ID Networkers

R Basic
HackerRank

R Intermediate
HackerRank

SQL Basic
HackerRank

SQL Intermediate
HackerRank

Go Programming Language Basic
HackerRank